



The 'sopra-paper' class

Documentation for the 'sopra-paper' class, version 1.0.07

13. December 2019

Prepared by:

Florian Sihler (florian.sihler@uni-ulm.de)

Description:

This documentation gives a comprehensive insight into the use of the standalone 'sopra-paper' class.

NO TIME TO SPY

1 General

1.1 Why, what for, and wherefore?

This $\text{\LaTeX} 2_{\epsilon}$ document class was written as part of the Sopra in the winter semester 2019 and summer semester 2020 and serves as the basis for the look of all documents of Team 20. The layout can act both as a classic book as well as an article (see: `usebook (useartcl) \rightarrow p. 5`). For structural reasons, this documentation was prepared in the latter, so no titles are placed; the header adapts accordingly to this circumstance!

To visualize the individual code snippets, the `sopra-listings` package is used. It is not relevant for the use of the document class.

The corresponding class should also be embedded in this document: `sopra-paper.cls`.

1.2 Dependencies

This package is based on the KOMA-Script classes `scrbook`⁽¹⁾ or `scrartcl`⁽²⁾ (see: `usebook (useartcl) \rightarrow p. 5`) as well as the following packages (all passed arguments are also listed, whereby an asterisk is set in the case of a value being passed. An exclamation mark before an “argument” declares that this is an option that controls the loading of the package):

- `tikz`
- `fontenc`^(T1)
- `inputenc`^(utf8)
- `babel`^(english,main=ngerman)
- `geometry`^(margin*, a4paper,bottom*)
- `CormorantGaramondrmfont!`
- `sfmathsffont!`
- `montserratsffont,defaultfam,*!`
- `lmodern`
- `microtype`
- `scrlayer-scrpage`
- `titlesec`
- `enumitem`
- `nowidow`^(all)
- `hyperref`^(hidelinks)

All of these packages should be part of a common $\text{\LaTeX}$ distribution (furthermore the TikZ library `calc` is also used).

1.3 Installation

The class is not distributed as a `.dtx` file, which is why the following options are available:

- The document class can be placed in the same directory as the document. In this case, the inclusion instruction reads:

```
\documentclass{sopra-paper}
```

- The document class can be placed in a subdirectory / in a directory shipped with the document. In this case, it is specified via the (relative) path:

```
\documentclass{./Mein/Pfad/zu/sopra-paper}
```

⁽¹⁾<https://www.ctan.org/pkg/scrbook>

⁽²⁾<https://www.ctan.org/pkg/scrartcl>

- The class can be placed (using a symlink or something similar) into your own texmf tree. For example, on Linux using texlive the class can be placed here: `~/texmf/tex/latex/`. The directory can be created and then updated using `texhash ~/texmf`. Now the class can be used like any other installed class:

```
\documentclass{sopra-paper}
```

Note: for this purpose there is a Python3.5 script called `installer.py` which, when placed in the folder above, can take care of the installation/update.

1.4 Further Particularities

In version 1.0.07 (`\thesopversion` $\rightarrow$ p. 7) there are no further particularities.

2 Class Configuration

2.1 Accepted Parameters

Like most, the document class accepts arguments. In addition to the arguments accepted for `article`, the ones listed below can be used, whereby for arguments with a “counter” option the one active by default is written first and the other in parentheses. Implicitly, this means:

```
\documentclass[final,nopar,sffont,defaultmode,showmail,dotoc,usebook,setstyle]%
{sopra-paper}
```

is called. While this document is invoked with:

```
\documentclass[useartcl,notoc]{sopra-paper}
```

which uses the `scrartcl` class and (at least automatically) does not generate a table of contents.

▷ `final` (draft)

In `draft` mode, some elements such as graphics cannot be typeset, in order to speed up the compilation process. This option is likewise accepted by many other packages as well.

▷ `nopar` (dopar)

Using `dopar` we set the indentation of a paragraph to the default value set by $\text{\LaTeX}$.

▷ `sffont` (rmfont)

Using `rmfont`, the font is set back to Roman.

- ▷ `defaultmode` (`print`)

Using `print`, the colors are reduced or, in parts, completely removed (converted to grayscale) to achieve a better result when printed.

- ▷ `showmail` (`hidemail`)

With `hidemail`, the email address is not shown when listing the authors.

- ▷ `dotoc` (`notoc`)

With `notoc`, the automatic typesetting of a table of contents is disabled.

- ▷ `usebook` (`useartcl`)

Controls whether `scrbook`⁽³⁾ (`usebook`) or `scrartcl`⁽⁴⁾ (`useartcl`) is used to typeset the document. With `scrartcl`, no chapters are available! Furthermore, the header changes in that the individual sections are now tracked. If the format is to be changed after it has already been generated differently once, the `<Dokumentname>.aux` and `<Dokumentname>.toc` files must be deleted!

- ▷ `setstyle` (`nosetstyle`)

With `setstyle`, it is automatically checked which packages of the `sopra-collection`⁽⁵⁾ are loaded, and the corresponding styles are set to produce a consistent layout.

2.2 Further Information

Work is currently underway on an adequate layout for presentations.

3 Command Overview

As a base class, the class does not add a large set of commands for the user. This task is left to the extending packages.

3.1 Setting Data

The following commands should be set in the preamble and also configure the metadata of the respective document (i.e. the document properties). They can be overwritten as often as desired (up to the point at

⁽³⁾<https://www.ctan.org/pkg/scrbook>

⁽⁴⁾<https://www.ctan.org/pkg/scrartcl>

⁽⁵⁾<https://github.com/EagleoutIce/sopra-collection>

which the document begins). Developer note: all these fields are available via `\sop@register@⟨Name⟩`, or their 'short' variants via `\sop@register@short@⟨Name⟩`.

▷ `\title[short]{NewTitle}`

Sets the title of the document to NewTitle.

▷ `\subtitle[short]{NewTitle}`

Sets the subtitle of the document to NewTitle.

▷ `\brief[short]{ShortDescription}`

Sets the short description of the document to ShortDescription. It is shown at the bottom left of the title page.

▷ `\authors[short]{AuthorList}`

This can be used to set the list of authors. It is recommended to use `\addAuthor`^{→ p. 7} for this.

▷ `\duedate[short]{YYYY-MM-DD}`

Sets the date of the document. `\DateConvert`^{→ p. 7} is used for the conversion.

▷ `\supervisor[short]{Name}`

Sets the supervisor to Name. This data field is not used in the base class.

▷ `\keywords[short]{Keywords}`

Sets the keywords for the metadata to Keywords.

▷ `\group[short]{Name}`

Sets the group of the respective author to Name.

▷ `\groupnum[short]{Number}`

In case of a numbering (or symbol, etc.) of the group, sets this to Number.

3.2 Helpful When Setting Data

▷ `\addAuthor{Name (Email)}`

Adds an author to the list of authors (`\authors`^{→p. 6}). For example: `\addAuthor{Florian Sihler (florian.sihler@uni-ulm.de)}`.

▷ `\setTeam{Name Number}`

Allows `\group`^{→p. 6} and `\groupnum`^{→p. 6} to be set in a single call. For example: `\setTeam{Affenbande 42}` or `\setTeam{{Mega Team} A}`.

▷ `\sopDisableTocRegister`

Disables the recording of sections (in the header).

▷ `\sopEnableTocRegister`

Enables the recording of sections (in the header).

3.3 Displaying the Data

▷ `\typesetAuthors`

Typesets the authors according to `showmail` with or without email address, so `\typesetAuthors` yields: Florian Sihler (florian.sihler@uni-ulm.de). With multiple authors, commas and the “and” are set correctly.

▷ `\thesopversion`

Outputs the current version of the `sopra-paper` document class. Note: via `\value{sopversion}` the version can be obtained as a 4-digit number: 1007.

▷ `\DateConvert{YYYY-MM-DD}`

Converts a date into a nicer form, thus:

`\DateConvert{2019-12-02}` % → 02. December 2019

3.4 General Formatting Commands

▷ `\imptext{Text}`

Highlights Text as **very important**.

▷ `\notetext{Text}`

Sets Text as an incidental note.

3.5 Colors

The following colors are available: `cprimary` ●, `cakzent` ●, `clight` ●, `cimportant` ●, `cwhite` ○ and `chighlight` ●.

A Related Classes and Packages

In the course of this $\text{\LaTeX 2}_{\epsilon}$ document class, a few other packages have also come into being. As of the current build, the following exist (packages and classes without a link are included alongside this one and can be found here: <https://github.com/EagleoutIce/sopra-collection>):

- `eagle-maps`^{[\(6\)](#)}
- `LILLY-Framework`^{[\(7\)](#)}
- `sopra-documentation`
- `sopra-models`
- `sopra-listings`
- `sopra-requirements`
- `sopra-attachments`

^{[\(6\)](#)}<https://github.com/EagleoutIce/eagle-maps>

^{[\(7\)](#)}<https://github.com/EagleoutIce/LILLY>